

Assignment 5

Artificial Intelligence

Q1. Minimax Search Algorithm (Case Study)

A gaming company is designing an AI for a two-player strategy game similar to chess. The AI must always choose the optimal move assuming the opponent is also playing optimally.

Tasks:

- Explain the Minimax algorithm and how it works in adversarial search.
- Represent the game as a game tree and show how Minimax evaluates nodes.
- Discuss the role of utility functions and depth limits.
- Analyze time and space complexity.
- Suggest improvements for handling large game trees.

Q2. Alpha-Beta Pruning (Case Study)

The same gaming company wants to improve the performance of its AI because Minimax is too slow for deep searches.

Tasks:

- Explain how Alpha-Beta Pruning optimizes Minimax.
- Illustrate pruning with a suitable example tree.
- Compare performance with and without pruning.
- Discuss best-case and worst-case scenarios.
- Explain how move ordering affects efficiency.

Q3. Expert System (Case Study)

A hospital wants to build an AI-based system that can assist doctors in diagnosing diseases based on symptoms and patient history.

Tasks:

- Define an Expert System and its components (knowledge base, inference engine, user interface).
- Explain rule-based reasoning (forward and backward chaining).
- Design a simple rule-based system for disease diagnosis.
- Discuss advantages and limitations of expert systems in healthcare.
- Suggest improvements using modern AI techniques.

Q4. Swarm Intelligence System (Case Study)

A smart city project aims to optimize traffic flow using decentralized decision-making inspired by nature.

Tasks:

- Define Swarm Intelligence and its characteristics.
- Explain how decentralized agents can solve complex problems.

Assignment 5

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- Compare swarm intelligence with traditional centralized systems.
- Provide examples such as particle swarm optimization or bird flocking.
- Discuss real-world applications in traffic and robotics.

Q5. Ant Colony System (Case Study)

A delivery company wants to find the shortest route for multiple delivery vehicles across a city with dynamic traffic conditions.

Tasks:

- Explain the Ant Colony Optimization (ACO) algorithm.
- Describe how ants use pheromone trails to find optimal paths.
- Model the delivery problem using ACO.
- Compare ACO with traditional algorithms like Dijkstra.
- Discuss advantages, limitations, and real-world applications.